English

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Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

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(Academic Session : 2024 - 2025)

NEET(UG)
MAJOR
28-11-2024

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE - MEA, B, C, D, L, M, N, O, P & Q

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

Test Booklet Code

E2

This Booklet contains **24** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are 720.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
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6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
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Name of the Candidate (in Capitals) : _____

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Candidate's Signature : _____ **Invigilator's Signature :** _____

Your Target is to secure Good Rank in Pre-Medical 2025

SUBJECT : PHYSICS

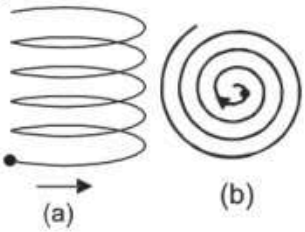
Topic : SYLLABUS-2

1. A motorcycle is going on an overbridge of radius R. The driver maintains a constant speed. As the motorcycle is ascending on the overbridge, the normal force on it

(1) increases (2) decreases

(3) remains the same (4) fluctuates

2. A particle is going with constant speed along a uniform helical and spiral path separately as shown in figure



(1) The velocity of the particle is constant in both cases

(2) The magnitude of acceleration of the particle is constant in both cases

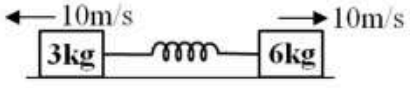
(3) The magnitude of acceleration is constant in (a) and decreasing in (b)

(4) The magnitude of acceleration is decreasing continuously in both the cases

3. Two particles of mass 2 kg and 1 kg are moving along the same straight line with speeds 2 m/s and 5 m/s respectively. The speed of COM of system when particles are moving in opposite directions is V_1 and when particles are moving in same direction is V_2. The ratio V_1/V_2 is :

(1) 3:4 (2) 4:3 (3) 9:1 (4) 1:9

4. In given diagram find out maximum energy stored in spring. Extension in spring is zero initially.



(1) 400 J (2) 300 J

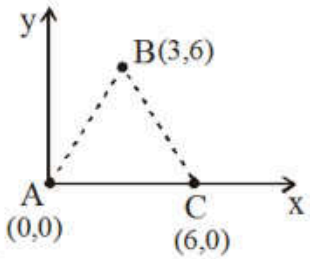
(3) 200 J (4) 100 J

5. Mass is non-uniformly distributed on the circumference of a ring of radius r and centre at origin. If a is the distance of centre of mass of ring from origin then :-

(1) $a = r$ (2) $a > r$

(3) $a < r$ (4) $o \le a \le r$

6. Consider the system of 3 point masses A,B and C as shown. Match entries of List-II for location of center of mass of system for each configuration of masses in List-I.



	List-I		List-II
(P)	$m_A = m_B = m_C$	(1)	$x_{CM} = 3$
(Q)	$m_A = m_C = 2m_B$	(2)	$x_{CM} = \frac{18}{5}$
(R)	$m_A = m_B = 2m_C$	(3)	$x_{CM} = 4$
(S)	$m_B = m_C = 2m_A$	(4)	$x_{CM} = \frac{12}{5}$

(1) P ⟶ 1;Q ⟶ 1;R ⟶ 2;S ⟶ 4

(2) P ⟶ 1;Q ⟶ 1;R ⟶ 4;S ⟶ 2

(3) P ⟶ 3;Q ⟶ 1;R ⟶ 5;S ⟶ 3

(4) P ⟶ 1;Q ⟶ 4;R ⟶ 5;S ⟶ 5

7. A bomb initially at rest explodes by itself into three equal mass fragments. The velocities of two fragments are $(3\hat{i} + 2\hat{j})$ m/s and $(-\hat{i} - 4\hat{j})$ m/s. The velocity of the third fragment is (in m/s) :-

(1) $2\hat{i} + 2\hat{j}$

(2) $2\hat{i} - 2\hat{j}$

(3) $-2\hat{i} + 2\hat{j}$

(4) $-2\hat{i} - 2\hat{j}$

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8. Two point masses m_1 and m_2 at rest in gravity free space are released from distance d. Find the velocity of centre of mass of system at the time of collision of masses:

(1) zero (2) 2 (3) 3 (4) None

9. The ratio of the radius of a planet 'A' to that of planet 'B' is 'r'. The ratio of acceleration due to gravity on the planets is 'x'. The ratio of the escape velocities from the two planets is :

(1) xr (2) $\sqrt{\frac{r}{x}}$ (3) $\sqrt{rx}$ (4) $\sqrt{\frac{x}{r}}$

10. The orbital velocity of an artificial satellite in a circular orbit very close to earth is V. The velocity of an another satellite orbitting in circular orbit at an altitude of 3R from earth's surface will be :-

(1) $\frac{V}{\sqrt{7}}$ (2) $\frac{V}{\sqrt{6}}$ (3) $\frac{V}{2}$ (4) $\frac{V}{\sqrt{2}}$

11. Two spheres of masses m and M are situated in air and the gravitational force between them is F. the space around the masses is now filled with a liquid of specific gravity 3. The gravitational force between bodies will now be :-

(1) F (2) $\frac{F}{3}$ (3) $\frac{F}{9}$ (4) 3F

12. Two concentric spherical shell have masses M_1 and M_2 and radii R_1 and R_2 ($R_2 > R_1$) then force on particle of mass m placed of a distance $\left(\frac{R_1+R_2}{2}\right)$ from centre of this system :-

(1) $\frac{4GM_2m}{(R_1+R_2)^2}$ (2) $\frac{4GM_1m}{(R_1+R_2)^2}$

(3) $\frac{GM_1m}{(R_1+R_2)^2}$ (4) Zero

13. A projectile is fired with a speed $v = 2\sqrt{gR}$ from the surface of earth. It escapes the gravitational pull of earth. Its speed in interstellar space will be:

(1) $\sqrt{gR}$ (2) $\sqrt{3gR}$

(3) $\sqrt{gR/2}$ (4) $\sqrt{2gR}$

14. By what amount, the energy of a satellite of mass m has to be increased to shift it from an orbit of radius r to orbit of radius $\frac{3}{2}r$? (mass of earth is M)

(1) $\frac{GMm}{r}$

(2) $\frac{GMm}{2r}$

(3) $\frac{GMm}{4r}$

(4) $\frac{GMm}{6r}$

15. Two point masses are placed at r distance away from each other. The ratio of their masses is 4 : 1. Then find ratio of gravitational force exerted by them on each other :-

(1) 1 : 2 (2) 2 : 1

(3) 4 : 1 (4) 1 : 1

16. In a gravitational field, the gravitational potential is given by $V = -\frac{K}{x}\frac{J}{kg}$, where K is a constant. The gravitational field intensity (in N/kg) at point (2, 0, 3) m is :-

(1) $+\frac{K}{2}$ (2) $-\frac{K}{2}$

(3) $-\frac{K}{4}$ (4) $+\frac{K}{4}$

17. **Statement-1 :** The formula $-GMm\left(\frac{1}{r_2}-\frac{1}{r_1}\right)$ is more accurate than the formula $mg(r_2 - r_1)$ for the difference of potential energy between two points r_2 and r_1 distance away from the centre of the earth.
Statement-2 : If the zero of potential energy is at infinity, the total energy of an orbiting satellite is negative of its kinetic energy.

(1) Both Statement-1 and 2 are correct

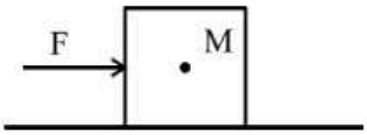
(2) Both Statement-1 and 2 are incorrect

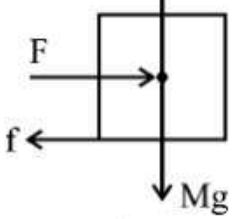
(3) Statement-1 is correct but Statement-2 is incorrect

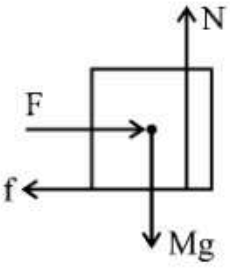
(4) Statement-1 is incorrect but Statement-2 is correct

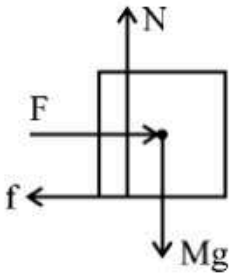
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18. If the block shown in the figure is in equilibrium, under the action of an external force. Then choose the correct free body diagram of the block :-



(1) 

(2) 

(3) 

(4) None of these

19. A ball is thrown on a horizontal surface in such a way that it initially slides with a speed v_0 without rolling. It gradually picks up rotation motion. Find the speed of the ball at which there will be rolling without slipping–

(1) $\frac{2}{7}v_0$ (2) $\frac{2}{5}v_0$

(3) $\frac{5}{7}v_0$ (4) $\frac{3}{5}v_0$

20. A body is rolling down an inclined plane. If kinetic energy of rotation is 40% of kinetic energy in translatory state, then the body is a :-

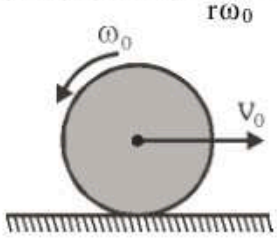
(1) Ring (2) Cylinder

(3) Hollow ball (4) Solid ball

21. When a solid sphere rolls without slipping down an inclined plane making an angle θ with the horizontal, the acceleration of its centre of mass is a. If the same sphere slides without friction, its acceleration is :-

(1) $\frac{7}{2}a$ (2) $\frac{5}{7}a$ (3) $\frac{7}{5}a$ (4) $\frac{5}{2}a$

22. A uniform circular disc placed on a rough horizontal surface has initial velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is :-



(1) 1/2 (2) 1 (3) 3/2 (4) 2

23. A uniform thick walled cylinder, a uniform thin walled cylinder, and a uniform solid cylinder all have the same mass and outer radius. Rank them according to decreasing (greatest to least) moment of inertia.



(1) Thick, thin, solid (2) Solid, thick, thin

(3) Thin, thick, solid (4) Solid, thin, thick

24. A circular disc has a mass of 1 kg and radius 40 cm. It is rotating about an axis passing through its centre and prependicular to its plane with a speed of 10 rev/s. The work done in joules in stopping it would be :-

(1) 4 (2) 47.5 (3) 79 (4) 158

25. Consider the following statements and choose the correct option among the following.

Statement-A: In pure translational motion at any instant of time, all particles of the body have the same velocity.

Statement-B: In pure rotation of the rigid body about a fixed axis, every particle of the body moves in a circle, which lies in a plane perpendicular to the axis and has its centre on the axis.

(1) Both A & B are correct

(2) A is correct and B is incorrect

(3) A is incorrect and B is correct

(4) Neither A is correct nor B is correct

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26. Two wires are made of same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 6A. If on applying a force F, length of first wire increases by $\Delta \ell$, how much force is required to stretch the second wire by $2\Delta \ell$?

(1) 9 F

(2) 36 F

(3) 72 F

(4) F

27. The upper end of a wire 1 m long and 4 mm radius is clamped. The lower end is twisted by an angle of 30°. The angle of shear is :-

(1) 12°

(2) 1.2°

(3) 0.12°

(4) 0.012°

28. In a tube, flow speed of water becomes double at some cross section as compare to first. The radius of new cross section will be :-

(1) Half of first

(2) 70% of first

(3) Double of first

(4) Same as first

29. Two water pipes of diameters 2 cm and 4 cm are connected with the main supply line. The velocity of flow of water in the pipe of 2 cm diameter is :-

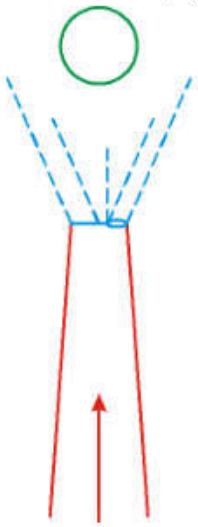
(1) 4 times that in the other pipe

(2) $\frac{1}{4}$ times that in the other pipe

(3) 2 times that in the other pipe

(4) $\frac{1}{2}$ times that in the other pipe

30. **Assertion** : A light celluloid ball placed in a stream of gas or water issuing at a high velocity from a tube with a narrow neck, the ball floats freely however in this stream (Fig)



Reason : The gas in the stream has a high velocity, the pressure inside the stream is above atmospheric.

(1) Assertion is true, Reason true and Reason is a correct explanation for Assertion

(2) Assertion is true, Reason is true, Reason is not a correct explanation for Assertion

(3) Assertion is true; Reason is false

(4) Assertion is false; Reason is true.

31. The shear stress at a point in a liquid is found to be 0.03 N/m^2. The velocity gradient at the point is 0.15 s^{-1}. Then its coefficient of viscosity is :

(1) 20 Poise (2) 2 Poise

(3) 0.2 Poise (4) 0.5 Poise

32. A wire of area of cross-section 10^{-6} m^2 is increased in length by 0.1%. The tension produced is 1000 N. The Young's modulus of the wire is :-

(1) 10^{12} N/m^2

(2) 10^{11} N/m^2

(3) 10^{10} N/m^2

(4) 10^{9} N/m^2

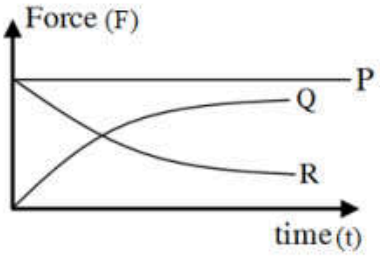
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33. A spherical ball of density (ρ) is dropped in a long column of viscous liquid of density (σ). Assuming $\rho > \sigma$, select the appropriate graph of which represent variation of magnitude of force w.r.t time for following statements :

A. gravitational force with time

B. viscous force with time

C. net force with time



(1) Q, R, P

(2) R, Q, P

(3) P, Q, R

(4) P, R, Q

34. Water rises to a height of 8 cm, above outside level when a long clean capillary tube is dipped in water having surface tension 7×10^{-2} Nm^{-1}. The diameter of capillary will be :

(1) 0.35 mm (2) 0.40 mm

(3) 0.50 mm (4) 1 mm

35. **Assertion (A) :** In Searle's experiment to find the Young's modulus of a given wire, it is advised to wait for few minutes before taking reading after changing load on wire.

Reason (R) : During this time wire gets free from kinks.

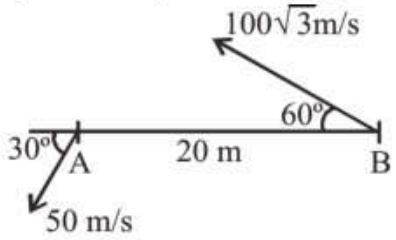
(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect.

(4) Assertion is incorrect but reason is correct.

36. Velocity of two particles A and B separated by distance 20 m is as shown in figure then what will be the angular velocity of B with respect to A



(1) 4.75 rad/s (2) 8.75 rad/s

(3) 16.75 rad/s (4) None of these

37. What is the position vector of a point mass moving on a circular path of radius of 10 m with angular frequency of 2 $rads^{-1}$ after $\pi/8$ s ? (the centre of the circle is (0, 0) and it starts from y-axis in clockwise direction).

(1) $5 . (\hat{i} + \hat{j})$

(2) $5\sqrt{2}\,(\hat{i} + \hat{j})$

(3) $\hat{i} + \hat{j}$

(4) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$

38. A particle starts circular motion with tangential acceleration $2m/s^2$. If radius of path is 2m then find out angle between net acceleration and velocity after 2s.

(1) $\tan^{-1}(2)$

(2) $\tan^{-1}(3)$

(3) $\tan^{-1}(4)$

(4) $\tan^{-1}(1/4)$

39. Keeping the angle of banking same, if the radius of curvature is made four times, the percentage increase in the maximum speed with which a vehicle can travel on a circular road is :-

(1) 25% (2) 50%

(3) 75% (4) 100%

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40. A particle of mass m collides head-on with another stationary particle of mass M such that the second particle starts moving and the first particle comes to rest after the collision. Which of the following conditions is valid if the coefficient of restitution is e?

(1) $e=0$

(2) $e = \frac{m}{M} \leqslant 1$

(3) $e = \frac{m}{M} \geqslant 1$

(4) $e = \frac{M}{m} \leqslant 1$

41. The time period of an earth satellite in a circular orbit of radius R is 2 days and its orbital velocity is v_0. If time period of another satellite in a circular orbit is 16 days then :-

(1) Its radius of orbit is 4R and orbital velocity is v_o

(2) Its radius of orbit is 4R and orbital velocity is $\frac{v_o}{2}$

(3) Its radius of orbit is 2R and orbital velocity is v_0.

(4) Its radius of orbit is 2R and orbital velocity is $\frac{v_o}{2}$

42. ***Assertion (A) :-*** A ladder is more apt to slip, when you are high up on it than when you just begin to climb.

Reason (R) :- When at high up a ladder, torque of gravity force about bottom point & friction required between ground and ladder is larger than he begin to climb.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

43. The density of gold is ρ and its bulk modulus is B. When pressure P is applied on piece of gold then its density will be ?

(1) $\frac{\rho P}{2B}$

(2) $\frac{\rho B}{2P}$

(3) $\frac{\rho B}{B-P}$

(4) $\frac{\rho P}{B-P}$

44. Water rises in a capillary tube to a height h. Choose the false statement regarding a capillary rise from the following.

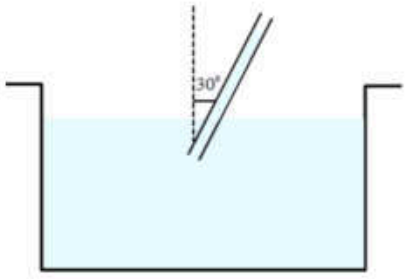
(1) On the surface of Jupiter, height will be less than h.

(2) In a lift moving up with constant acceleration, height is less than h.

(3) On the surface of the moon, the height is more than h.

(4) In a lift moving down with constant acceleration height is less than h.

45. A capillary of radius 0.5mm is dipped into a liquid of density 0.8 g cm^{-3} and surface tension 50 dyne cm^{-1}. Assuming angle of contact to be zero, the length of liquid column inside capillary above normal liquid level when the capillary is inclined at 30° to vertical will be



(1) 1.5cm

(2) 2.0cm

(3) 2.6cm

(4) 3.0cm

SUBJECT : CHEMISTRY

Topic : SYLLABUS-2

46. The ratio of minimum and maximum wavelengths of Lyman series in spectrum of Li^{+2} ion is :-

(1) 4 : 27 (2) 3 : 4 (3) 4 : 9 (4) 4 : 3

47. Which of the following is not a characteristic of planck's quantum theory of radiation ?

(1) Radiations are associated with energy

(2) Magnitude of energy associated with a quantum is equal to hν.

(3) Radiation energy is neither emitted nor absorbed continuously.

(4) A body can emit less or more than a quantum of energy.

48. Arrange the following in increasing order of energy :-

(i) $n = 4, \ell = 2, m = -1, s = +½$
(ii) $n = 3, \ell = 2, m = -1, s = -½$
(iii) $n = 4, \ell = 0, m = 0, s = +½$
(iv) $n = 5, \ell = 0, m = 0, s = -½$

(1) (i) < (ii) < (iii) < (iv)

(2) (iii) < (ii) < (iv) < (i)

(3) (iii) < (iv) < (ii) < (i)

(4) (ii) < (iii) < (i) < (iv)

49. The kinetic and potential energy of electron in 3rd shell of H-atom in eV are respectively :-

(1) −1.51, −3.02 (2) 1.51, −3.02

(3) −3.02, 1.51 (4) 1.51, −1.51

50. Which of 1s, 2p and 4f does not have any spherical node ?

(1) 1s only (2) 1s and 2p only

(3) 2p and 4f only (4) 1s, 2p and 4f all

51. For $n + \ell = 4$, calculate number of orbitals :-

(1) 1 (2) 3 (3) 4 (4) 9

52. Which of the following is not a representative element :-

(1) Tellurium (2) Tantalum

(3) Thallium (4) Astatine

53. IUPAC symbol of the element which belong to IVB group and 7th period :-

(1) Uth (2) Uqn

(3) Unq (4) Upn

54. Electronic configuration of Th (Thorium) is :

(1) $[Rn]\ 5f^2\ 6d^2\ 7s^2$ (2) $[Rn]\ 5f^0\ 6d^2\ 7s^2$

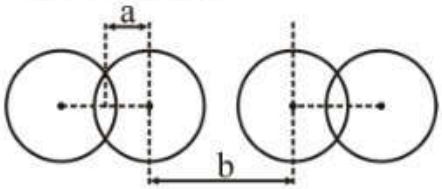
(3) $[Rn]\ 5f^3\ 6d^2\ 7s^2$ (4) $[Rn]\ 5f^3\ 6d^1\ 7s^2$

55. The relative penetration power of different subshells is :

(1) s > p > d > f (2) s < p < d < f

(3) s < d < p < f (4) d < s < p < f

56. Which type of atomic radius represented by 'a' & 'b' in the given diagram :-



(1) a = Atomic radius, b = Metallic radius

(2) a = Covalent radius, b = Vander Waal's radius

(3) a = Ionic radius, b = Covalent radius

(4) a = Covalent radius, b = Atomic radius

57. Size of Mo is very similar to W due to -

(1) Difference of atomic number by one

(2) Contraction in size in first transition series element.

(3) Lanthanide contraction

(4) actinide contraction

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58. ***Assertion :*** Size of anion is larger than its parent atom.
Reason : Z_{eff} of anion is greater than that of their parent atom.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

59. Highest ionisation energy and minimum ionisation energy in periodic table is of respectively :

(1) F, Fr

(2) He, Cs

(3) He, Rb

(4) Rn, Li

60. **Assertion :** Electron affinity of F is greater than Cl.
Reason : Ionisation energy of F is less than Cl.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

61. Element having positive value of electron gain enthalpy is :-

(1) Na (2) C (3) N (4) O

62. Which of the following strongly absorbs CO_2 ?

(1) K_2O

(2) BeO

(3) P_4O_{10}

(4) conc. H_2SO_4

63. Correct order of acidic strength :

(1) $CO_2 > N_2O_5 > SO_3$

(2) $Cl_2O_7 > SO_3 > P_4O_{10}$

(3) $Na_2O > MgO > Al_2O_3$

(4) $K_2O > CaO > MgO$

64. **Assertion :** In boron family, down the group electronegativity regularly decreases.
Reason : Atomic size of Ga is nearly equal to size of Al.

(1) Assertion and Reason both are correct but Reason is not correct explanation of Assertion.

(2) Assertion is incorrect but Reason is correct.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion and Reason both are correct and Reason is correct explanation of Assertion.

65. **Assertion :-** H_2Se is more acidic than H_2S.
Reason :- S is more electronegative than Se.

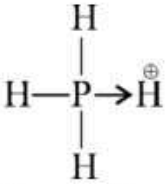
(1) Both Assertion and reason are correct and reason is correct explanation for assertion.

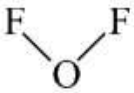
(2) Both Assertion and reason are correct and reason is not the correct explanation for assertion.

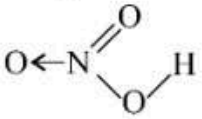
(3) Assertion is correct but Reason is incorrect.

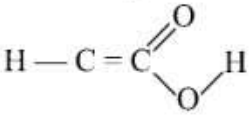
(4) Assertion is incorrect but Reason is correct.

66. Which formula does not correctly represent the bonding capacity of the atom involved

(1) 

(2) 

(3) 

(4) 

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67. Which of the following is most favourable condition for bond formation ?

(1) Minimum potential energy and minimum attractive force

(2) Maximum potential energy and maximum attractive force

(3) Minimum potential energy and maximum attractive force

(4) Maximum potential energy and minimum attractive force

68. Octet rule is not valid for the molecule :-

(1) CO_2 (2) NO

(3) O_2 (4) PCl_3

69. **Assertion :-** In NH_3, N is sp^3 hybridised but angle is found to be 107°.

Reason :- The decrease in bond angle is due to repulsion by the lone pair.

(1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.

(2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion

(3) Assertion is correct, reason is incorrect

(4) Assertion is incorrect, reason is correct.

70. Select the incorrect order of bond energy :-

(1) $CO > N_2$

(2) $HF > HI$

(3) $F_2 > HF$

(4) $H_2 > F_2$

71. In Lewis structure of ozone (O_3), formal charges on all three oxygen atoms can be ?

(1) 0, 0, 0

(2) +1, −1, 0

(3) +1, +1, +1

(4) +1, 0, +1

72. **Statement-1 :** Benzene itself is non-polar, but on monosubsitution it becomes polar.

Statement-2 : Dipole moment of bisubsituted benzene is always zero.

(1) Both Statement-1 and Statement-2 are incorrect.

(2) Statement-1 is correct but Statement-2 is incorrect.

(3) Statement-1 is incorrect but Statement-2 is correct.

(4) Both Statement-1 and Statement-2 are correct.

73. Hydrogen bonds are formed in many compounds e.g., H_2O, HF, NH_3. The boiling point of such compounds depends to a large extent on the strength of hydrogen bond and the number of hydrogen bonds. The correct decreasing order of the boiling points of above compounds is :

(1) $HF > H_2O > NH_3$

(2) $H_2O > HF > NH_3$

(3) $NH_3 > HF > H_2O$

(4) $NH_3 > H_2O > HF$

74. The oxygen molecule is paramagnetic because

(1) the bonding electrons number is higher than antibonding electrons in the molecular orbital

(2) it contains unpaired electron in the antibonding molecular orbitals

(3) it contains unpaired electrons in the bonding molecular orbitals

(4) the number of bonding electrons equals that of the antibonding electrons in the molecular orbitals

75. Stability of the species Li_2, Li_2^- and Li_2^+ increases in the order of :-

(1) $Li_2 < Li_2^+ < Li_2^-$

(2) $Li_2^- < Li_2^+ < Li_2$

(3) $Li_2 < Li_2^- < Li_2^+$

(4) $Li_2^- < Li_2 < Li_2^+$

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76. **Assertion :** Generally ionic compounds have high boiling point and melting point.

Reason : Due to strong electrostatic force of attraction among oppositely charged ions.

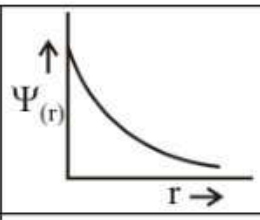
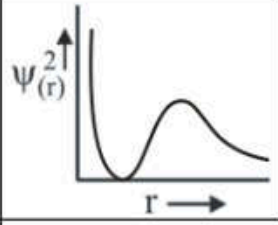
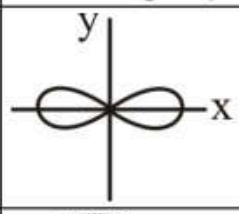
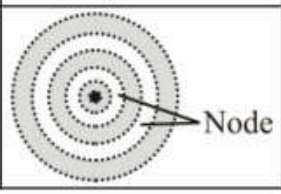
(1) Assertion and Reason both are correct and reason is correct explanation of Assertion.

(2) Assertion and Reason both are correct but reason is not correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

77. Match the column-I with column-II :

Column-I		Column-II	
(P)		(A)	3s
(Q)		(B)	2s
(R)		(C)	2p
(S)		(D)	1s

(1) (P) - D, (Q) - B, (R) - C, (S) - A

(2) (P) - A, (Q) - C, (R) - B, (S) - D

(3) (P) - D, (Q) - C, (R) - B, (S) - A

(4) (P) - A, (Q) - B, (R) - C, (S) - D

78. Which of the following statement is incorrect :

(1) The effect of Heisenberg uncertainty principle is significant only for motion of microscopic objects and is negligible for that of macroscopic objects.

(2) In multi-electron atom, energies of the orbitals in the same subshell decrease with increase in the atomic number.

(3) 2s subshell have 2 spherical nodes.

(4) The negative electronic energy for hydrogen atom means that the energy of the electron in H-atom is lower than the energy of a free electron at rest.

79. Potential energy of electron present in He^+ is :-

(1) $\frac{2Ke^2}{r}$

(2) $\frac{-2Ke^2}{r^2}$

(3) $\frac{-2Ke^2}{r}$

(4) $\frac{-4Ke^2}{r}$

80. Incorrect match is :

(1) Natural transuranic element ⇒ Np, Pu

(2) Rare earth metals ⇒ Am, Cm

(3) Diagonally related elements ⇒ Be, Al

(4) Typical elements ⇒ Mg, Al

81. **Assertion :-** Group 17 elements (Halogens) have very high negative electron gain enthalpies.

Reason :- They can attain stable noble gas electronic configuration by picking up an electron.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

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82. The nomenclature of ICl is iodine chloride, because :-

(1) Size of I < Size of Cl

(2) Atomic number of I > Atomic number of Cl

(3) E.N. of I < E.N. of Cl

(4) E.A. of I < E.A. of Cl

83. Find the total no. of species in which dz^2 orbital participate in the hybridisation :

I_3^+, PCl_5, IF_7, NH_4^+, CH_4, CO_2, XeF_4

(1) 3 (2) 4

(3) 5 (4) None of these

84. **Assertion :** 2nd period elements do not expand their octet.

Reason : 2nd period elements do not have vacant 2d-orbitals.

(1) Assertion is true but the reason is false

(2) Assertion is false but reason is true

(3) Both assertion and reason are true and the reason is the correct explanation of assertion

(4) Both assertion and reason are true but reason is not the correct explanation of assertion

85. Which of the following statement regarding NH_3 and NF_3 is correct :

(1) NH_3 has pyramidal and NF_3 has trigonal planar shape

(2) Bond angle in NH_3 is smaller than NF_3

(3) Resultant dipole moment of NH_3 is $(4.90\times10^{-30}Cm)$ and that of NF_3 is $(0.8\times10^{-30}Cm)$

(4) They both are sp^2 hybridised

86. For true statement use 'T' for false use 'F' :-

(a) XeF_4 polar as well as planar.

(b) Bond length order $\Rightarrow$ Si – O > Cl – O

(c) In urea inter molecular H-bonding is not possible

(d) Covalent character :- $SiCl_4$ > CuCl > NaCl

(e) Melting point $\Rightarrow$ HI > HBr > HCl

(1) F, T, F, F, F (2) T, F, F, T, T

(3) F, T, F, T, T (4) T, F, T, F, T

87. **Assertion (A) :** Bond order of N_2^+ and N_2^- are same

Reason (R) : N_2^+ is more stable than N_2^-

(1) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

88. Correct order of spin only magnetic moment (in BM) is :

(1) $O_2 = B_2$ (2) $O_2^{\oplus} = N_2^{\ominus}$

(3) Both (1) & (2) (4) $N_2 > O_2$

89. Polarisation is the distortion of the shapes of an anion by an adjacently placed cation. Which of the following statements is correct ?

(1) Maximum polarisation is brought about by a cation of high charge

(2) Minimum polarisation is brought about by a cation of low radius

(3) A large cation is likely to bring about a large degree of polarisation

(4) Polarising power of cation is less than that of anion

90. **Assertion :** At room temperature O_2 and N_2 are stable molecule but S_2 and P_2 are less stable.

Reason : $2p_\pi - 2p_\pi$ collateral overlapping is effective but $3p_\pi - 3p_\pi$ overlapping is weak.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not correct explanation of assertion.

(3) Assertion is correct, reason is incorrect

(4) Both assertion and reason are incorrect.

SUBJECT : BIOLOGY

Topic : SYLLABUS-2

91. **Statement-I :** The chloroplast contain chlorophyll and carotenoids which are responsible for trapping light energy essential for photosynthesis.
Statement-II : The matrix of mitochondria possess single, circular DNA molecule, few RNA molecules, ribosome and components required for protein synthesis.
In the light of above statements, choose most appropriate answer from the given options.

(1) Both Statements are correct.

(2) Both Statements are incorrect.

(3) Statement-I is correct and Statement-II are incorrect.

(4) Statement-I is incorrect and Statement-II are correct.

92. An elaborate network of filamentous proteinaceous structure present in the cytoplasm is collectively refered to :-

(1) Micro filament

(2) Intermediate filament

(3) Cytoskeleton

(4) Axoneme

93. Which cell organelle of endomembrane system contains hydrolytic enzymes ?

(1) Vacuole

(2) Golgibody

(3) Lysosome

(4) Endoplasmic reticulum

94. Select the incorrect statement-

(1) Interphase nucleus has highly extended and elaborate nucleoprotien fibres.

(2) Perinuclear space is 10 to 50 µm.

(3) During different stages of cell division, cells show structured chromsomes in place of the nucleus.

(4) Every chromosome essentially has centromere.

95. Which chromosome has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm ?

(1) Metacentric chromosome

(2) Sub-metacentric chromosome

(3) Acrocentric chromosome

(4) Telocentric chromosome

96. Which one of the following cell organelles is responsible for the formation of glycoproteins and glycolipids ?

(1) Mitochondria (2) Ribosomes

(3) Nucleus (4) Golgi apparatus

97. **Assertion (A) :** Golgi apparatus remains in close association with endoplasmic reticulum.
Reason (R) : Materials to be packaged in the form of vesicles, from the ER fuse with the trans face of the golgi apparatus and move towards the maturing face.

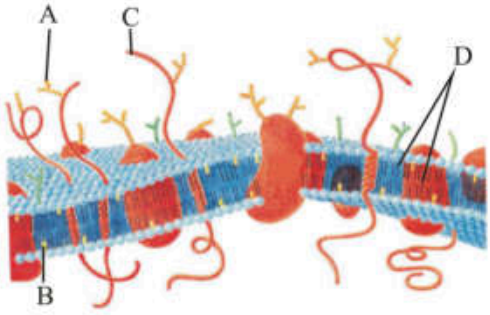
(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) (A) is correct but (R) is incorrect

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

98. Identify the A, B, C and D and select the correct match



(1) A - Sugar (2) B - Protein

(3) D - Cholesterol (4) C - Phospholipid

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99. Which of the following phenomenon occurs during telophase ?

(1) Spindle fibres attach to kinetochores of chromosomes.

(2) Centromeres split and chromatids separate

(3) Chromatids move to opposite poles.

(4) Nucleolus, Golgi complex and ER reform.

100. A very significant contribution of mitosis is-

(1) Cell repair

(2) Reduction in number of chromosomes

(3) Increase genetic variability

(4) Recombination

101. In which stage of cell division chromosomes show some dispersion but do not reach the extremely extended state of the interphase nucleus?

(1) Metaphase (2) Metaphase-I

(3) Telophase-I (4) Telophase

102. Separation of homologous chromosomes occur but centromere does not split in-

(1) Prophase-I (2) Metaphase-I

(3) Anaphase-I (4) Telophase-I

103. The complete disintegration of __A__ marks the start of __B__ of mitosis, hence the chromosomes are spread through the cytoplasm of the cell. Identify A and B.

(1) A–Cell membrane, B–Prophase

(2) A–Nuclear membrane, B–Prophase

(3) A–Nuclear membrane, B–Metaphase

(4) A–Cell membrane, B–Metaphase

104. Bivalent chromosomes align on equatorial plate at which stage?

(1) Prophase-I (2) Anaphase-I

(3) Metaphase-I (4) Telophase-I

105. Select the correct statements :

(a) Meiosis decreases the genetic variability in the population of organism from one generation to the next.

(b) Variations are very important for the process of evolution.

(c) Meiosis-II resembles a normal mitosis usually.

(d) Interkinesis is generally short lived.

(1) a, b and c only (2) b and c only

(3) b, c and d only (4) c and d only

106. Which of the following statement is incorrect ?

(1) The nuclear membrane and nucleolus reappear in telophase-I

(2) After telophase-I dyad of cells appear

(3) The stage between two meiotic divisions is called interphase

(4) Pairing of homologous chromosomes occur in zygotene

107. Match the following-

	Column-I		Column-II
(a)	Prophase-I	(i)	Bivalent chromosomes align on equatorial plate
(b)	Metaphase-I	(ii)	Homologous chromosome separate
(c)	Anaphase-I	(iii)	Synaptonemal complex formation
(d)	Telophase-I	(iv)	Nuclear membrane reappear

(1) (a–i), (b–ii), (c–iii), (d-iv)

(2) (a–iii), (b–ii), (c–iv), (d-i)

(3) (a–iii), (b–i), (c–ii), (d-iv)

(4) (a–iv), (b–iii), (c–ii), (d-i)

108. Which statements are not correct for meiosis?

(A) It is called the reductional division

(B) Meiosis occurs in the diploid cells

(C) Meiosis II is similar to mitosis

(D) A very significant contribution of meiosis is cell repair

(E) It also increases the genetic variability

(1) Both (B) and (D) (2) Only (D)

(3) (B), (C) and (E) (4) Only (E)

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109. Select the correct statements :
(a) Although cell growth is a continuous process, DNA synthesis occurs only during one specific stage in the cell cycle.
(b) Cells in the G_0 stage of cell cycle remain metabolically active.
(c) The chromosomal material becomes untangled during the process of chromatin condensation.
(d) Meiosis involves two sequential cycles of DNA replication.
(1) a, b and c (2) b, c and d
(3) a, b and d (4) a, c and d

110. Cytokinesis in plant cell achieved by ___A___ formation and cytokinesis in animal cell achieved by _B_ formation.
(1) A-Nuclear membrane B-Vacuole
(2) A-Furrow B-Nucleolus
(3) A-Cell plate B-Mitochondria
(4) A-Cell plate B-Furrow

111. **Statement-I :** All the elements present in a sample of earth crust are also present in a sample of living tissue.
Statement-II : Secondary metabolites have identifiable functions and play known roles in normal physiological processes.
Select the correct answer from given below options.
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct and Statement-II is incorrect.
(4) Statement-I is incorrect and Statement-II is correct.

112. Palmitic acid has ___ carbons including carboxyl carbon and arachidonic acid has ___ carbons atoms including carboxyl carbon.
Fill in the blanks respectively --
(1) 14 ; 18 (2) 16 ; 20 (3) 14 ; 16 (4) 16 ; 21

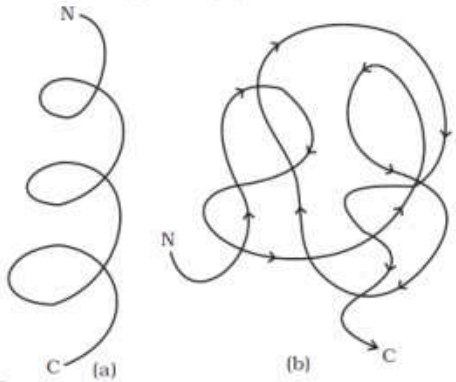
113. Match the column-I with column-II and choose correct option ?

	Column-I		Column-II
(a)	Alkaloids	(i)	Carotenoids
(b)	Lectins	(ii)	Morphin
(c)	Drugs	(iii)	Curcumin
(d)	Pigments	(iv)	Concanavalin-A

(1) a-i, b-ii, c-iii, d-iv (2) a-ii, b-iii, c-iv, d-i
(3) a-iii, b-iv, c-i, d-ii (4) a-ii, b-iv, c-iii, d-i

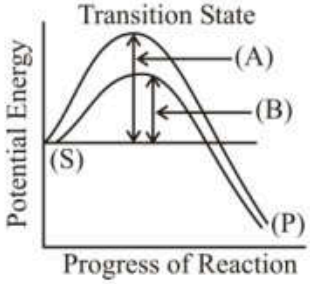
114. Which of the following statement is correct :-
(1) Living state is an equilibrium steady state
(2) Metabolic pathway is always circular
(3) Sum total of all the chemical reaction taking place in the body is known as catabolism
(4) Inulin is a polymer of fructose

115. Identify the configuration of polypeptide chain in given figure (a) and (b)



	(a)	(b)
(1)	Primary	Tertiary
(2)	Secondary	Tertiary
(3)	Secondary	Quaternary
(4)	Primary	Secondary

116. In the following diagram A and B is :



(1) (A) Activation energy without enzyme.
(B) Activation energy with enzyme.
(2) (A) Activation energy with enzyme.
(B) Activation energy without enzyme.
(3) (A) ⟶ Substrate
(B) ⟶ Product
(4) (A) ⟶ Product
(B) ⟶ Substrate

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117. Haem is the prosthetic group in :

(1) Peroxidase (2) Carboxypeptidase

(3) RuBisco (4) Carbonic anhydrase

118. $E + S \rightleftharpoons ES \longrightarrow EP \longrightarrow E + P$

Which of the following is correct regarding above reaction ?

(1) E-Enzyme, S-Product

(2) E-Enzyme, EP-Product

(3) S-Substrate, ES-Enzyme

(4) S-Substrate, P-Product

119. Study the following statements carefully and select the correct statement.

(1) The enzymes, catalysing inter-conversion of optical, geometric or positional isomers are called isomerases

(2) Enzymes catalysing the linking together of two compounds are grouped in ligases.

(3) Enzymes catalysing hydrolysis of substrates included in hydrolases

(4) All of the above

120. Which is the correct order of cell cycle ?

(1) $G_1 \longrightarrow S \longrightarrow G_2 \longrightarrow$ Prophase $\longrightarrow$ Metaphase $\longrightarrow$ Anaphase $\longrightarrow$ Telophase $\longrightarrow$ Cytokinesis

(2) $G_2 \longrightarrow G_1 \longrightarrow S \longrightarrow$ Prophase $\longrightarrow$ Metaphase $\longrightarrow$ Anaphase $\longrightarrow$ Telophase $\longrightarrow$ Cytokinesis

(3) $S \longrightarrow G_1 \longrightarrow G_2 \longrightarrow$ Prophase $\longrightarrow$ Metaphase $\longrightarrow$ Anaphase $\longrightarrow$ Telophase $\longrightarrow$ Cytokinesis

(4) Prophase $\longrightarrow$ Metaphase $\longrightarrow$ Anaphase $\longrightarrow$ Telophase $\longrightarrow$ Cytokinesis $\longrightarrow S \longrightarrow G_2 \longrightarrow G_1$

121. Match the following column :-

(1)	Human R.B.C.	(a)	egg of an ostrich
(2)	Largest isolated single cell	(b)	7 µm in diameter
(3)	Nerve cell	(c)	Longest cell
(4)	Bacteria	(d)	3 to 5 µm

(1) 1-d, 2-b, 3-c, 4-a (2) 1-b, 2-c, 3-d, 4-a

(3) 1-b, 2-a, 3-c, 4-d (4) 1-d, 2-b, 3-a, 4-c

122. Consider the following statements and choose the correct option :-

A. Larger and more numerous nuclei are present in cells actively carrying out lipid synthesis.

B. Chromatin net is made up of protein and nucleic acid.

C. The inner membrane of nuclear envelope usually remains continuous with the endoplasmic reticulum.

D. In case of acrocentric chromosome the centromere is situated close to it end forming one extremely short term.

(1) A, B and C (2) B, C and D

(3) A and C (4) B and D

123. In a diploid cell after S-phase quantity of DNA is 20 pico gram then after meiosis-II each daughter nucleus will be :-

(1) Diploid with 10 pico gram DNA

(2) Haploid with 10 pico gram DNA

(3) Haploid with 5 pico gram DNA

(4) Diploid with 5 pico gram DNA

124. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Anaphase-II begins with the simultaneous splitting of the centromere of each chromosome allowing than to move toward opposite poles of the cell.

Reason (R) : The shortening of microtubules attached to kinetochore takes place during anaphase-II.

Choose the correct answer from the options given below :

(1) Both (A) and (R) are true but (R) is not the correct explaination of (A).

(2) (A) is true but (R) is false.

(3) (A) is false but (R) is true.

(4) Both (A) and (R) are true and (R) is correct explaination of (A).

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125. Find out the incorrect statement.

(1) The protein portion of the enzymes is called, apoenzyme

(2) Catalytic activity is lost when Co-factor is removed from the enzyme.

(3) Co-factor do not play a crucial role in the catalytic activity of the enzyme.

(4) All of these

126. Exoskeleton of arthropods have a complex polysaccharide called _____.

(1) Glycogen (2) Hemicellulose

(3) Inuline (4) Chitin

127. Which one of the following is monosaccharide?

(1) Cellulose (2) Glycogen

(3) Glucose (4) Maltose

128. Molecule whose molecular weight do not exceed 800 dalton and come under acid insoluble fraction. Above statement is suitable for which biomolecule ?

(1) Protein (2) Polysaccharide

(3) Lipid (4) Nucleic acid

129. Given below are two statements-

Statement-I : In some organisms, Karyokinesis is not followed by cytokinesis.

Statement-II : Liquid endosperm of coconut have multinucleated condition.

Choose the correct answer :

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

130. How many of the following are not related to mitotic division ?

karyokinesis, kinetochore, metaphase plate, pachytene, crossing over, diplotene, spindle fiber, telophase.

(1) 1 (2) 2 (3) 3 (4) 5

131. Which of the following is true for telophase-I ?

(1) Nuclear membrane reappears

(2) Nucleolus disappears

(3) Chromatin condensed into chromosome

(4) Each chromosome has one chromatid

132. Telophase I results in the formation of –

(1) Two diploid cell (2) Two haploid cell

(3) Four haploid cell (4) Four diploid cell

133. In which phase centriole begins to move towards opposite pole of cell?

(1) Telophase (2) Anaphase

(3) Metaphase (4) Prophase

134. Choose the correct statement :

(1) In plant cells the vacuole can occupy more than 95% of volume of the cell usually.

(2) The vacuole is bounded by a single membrane called tonoplast.

(3) In *Amoeba* the contractile vacuole is important for engulfing

(4) In many cells, as in protista, food vacuole are formed for osmoregulation

135. DNA is not present in –

(1) Chloroplast (2) Ribosomes

(3) Nucleus (4) Mitochondria

136. In heavy smokers the alveoli of the lungs are enlarged and damaged which reduces the surface area of the exchange of respiratory gases. This condition is called :-

(1) Asthma (2) Silicosis

(3) Emphysema (4) Insomnia

137. If expiratory reserve volume is 1100 ml residual volume is 1200 ml and tidal volume is 500 ml, what shall be the functional residual capacity–

(1) 1600 ml (2) 2800 ml

(3) 2300 ml (4) 1200 m

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138. The structure which prevents the entry of food into respiratory tract is–

(1) Pharynx (2) Larynx

(3) Glottis (4) Epiglottis

139. Primary secondary, tertiary bronchi and initial bronchioles are supported by :

(1) Complete cartilaginous rings

(2) Incomplete cartilaginous rings

(3) Complete chitinous rings

(4) Incomplete chitinous rings

140. Thoracic chamber is formed dorsally by-

(1) Vertebral column

(2) Sternum

(3) Diaphragm

(4) Clavicle

141. What amount of carbon dioxide is delivered by every 100 ml of deoxygenated blood to the alveoli?

(1) 4 ml (2) 5 ml

(3) 15 ml (4) 20 ml

142. **Assertion (A) :-** The anatomical setup of lungs in thorax is such that any charge is the volume of thoracue cavity will be reflected in the pulmonary cavity.

Reason (R) :- Mammalian lungs are solid and spongy without muscle so power of self contration and relaxation is not present in mammalian lungs.

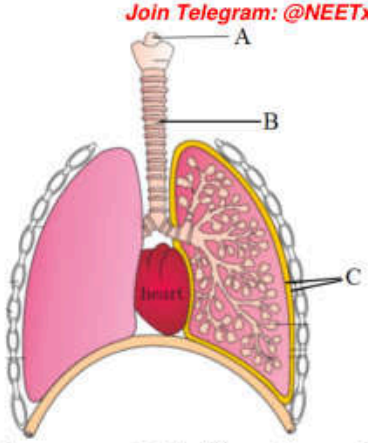
(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** and **Reason** are false.

143.



Choose the correct labelling from the given diagram :-

(1) A-Larynx, B-Plenural membrane, C-Trachea

(2) A-Trachea, B-larynx, C-Pleural membrane

(3) A-Epiglottis, B-Trachea, C- Pleural membrane

(4) A-Trachea, B-Plural membrane, C-Larynx

144. **Statement-I :-** Pressure contributed by mixture of gases is called partial pressure.

Statement-II :- High pO_2 and lesser H^+ concentration is favourable for the formation of oxyhaemoglobin in Alveoli.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) **Statement I** is correct but **Statement II** is incorrect.

145. Which of the following statement is correct ?

(1) O_2 is utilised by the organisms to directly break down simple molecules like glucose, amino acids, fatty acids etc.

(2) Earth worms use network of tubes (tracheal tubes) to transport atmospheric air within the body.

(3) Lungs are used by terrestrial organisms for the exchange of gases.

(4) Insects use their moist cuticle to transport atmospheric air within the body.

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146. Read the following statements :

(A) Normal heart rate is 72 betas/min.

(B) Cardiac cycle is complete in 8 sec.

(C) We can diagnose abnormality of heart by ECG.

(D) Cholesterol can form stones in kidney.

(E) RAAS increases BP.

(F) Aldosteron is secreted by JG cells.

Which of the following is correct statement ?

(1) A, C and E (2) A, B, C and E

(3) A, B, C, E and F (4) B, D and F

147. Find out incorrect statement ?

A. Conditional reabsorption of Na^+ and water takes place in distal convoluted tubule.

B. Atrial natriuretic factor cause vasodilation which causes increase in blood pressure.

C. An Adult human excrete on an average 1 to 1.5 litres of urea per day.

D. Inflammation of bowman's capsule of kidney is called glomerulonephritis.

(1) A & B only (2) B & C only

(3) B,C,D only (4) A,B,C,D

148. Malpighian tubules are excretory structures found in:

(1) Planaria (2) Prawn

(3) Earthworm (4) Cockroach

149. The thickness of each kidney of an adult human is about:

(1) 14-15 cm (2) 10-12 cm

(3) 21-25 cm (4) 2-3 cm

150. A minute vessel runs parallel to the Henle's loop of juxta medullary nephrons called :

(1) Efferent arteriole (2) Renal vein

(3) Vasa recta (4) Dorsal aorta

151. Which one is ammonotelic in nature :

(1) Aquatic insects (2) Bird

(3) Snake (4) Human

152. The cells named podocytes occur in

(1) Inner wall of Bowman's capsule

(2) Outer wall of Bowman's capsule

(3) Large intestine

(4) Neck region of nephrons

153. The structural and functional unit of kidney is :-

(1) Nephron (2) Neuron

(3) Neuroglia (4) Villi

154. Which layer of blood vessels is called endothelium ?

(1) Tunica intima

(2) Tunica media

(3) Tunica externa

(4) All of above

155. ___A___ grouping is based on the presence or absence of ___B___ surface antigen on the RBC :

(1) ABO, Three (2) Rh, Two

(3) ABO, Two (4) Rh, One

156. Lymphocyte constitute how much percentage of leucocytes

(1) 75% (2) 55%

(3) 30-32% (4) 20-25%

157. A patch of nodal tissue is present in the right upper corner of the right atrium called :-

(1) Purkinje fibres

(2) Sino-atrial node (SAN)

(3) Atrio-ventricular node (AVN)

(4) Ventricular node (VN)

158. Pulmonary artery carries :-

(1) Deoxygenated blood from heart to lungs

(2) Deoxygenated blood from lungs to heart

(3) Oxygenated blood from heart to lungs

(4) Oxygenated blood from lungs to heart

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159. The part of the heart that pumps blood to the lung is :-

(1) Right atria (2) Left ventricle

(3) Right ventricle (4) Left atria

160. Parasympathetic neural signals decrease the -

(1) Rate of heart beat

(2) Cardiac output

(3) Speed of conduction of action potential

(4) All of above

161. Match the column-I and II choose the correct option :

Column-I		Column-II	
(i)	Erythrocytes	(a)	Blood coagulation
(ii)	Leucocytes	(b)	1.5 lakh - 3.5 lakh.
(iii)	Platelets	(c)	Nucleated
(iv)	Fibrinogen	(d)	Life span 120 days

(1) i-c, ii-d, iii-a, iv-b (2) i-d, ii-c, iii-a, iv-b

(3) i-d, ii-c, iii-b, iv-a (4) i-b, ii-a, iii-c, iv-d

162. Rate of heart beat can be increased by:

(A) Sympathetic neural signals.

(B) Parasympathetic neeural signals.

(C) Adrenal medullary hormones.

(D) Vagus nerve.

(E) Thyroxine hormone

(F) Acetylcholine

(1) A, C, E (2) D, E, F

(3) A, C, D, (4) B, D, F

163. P wave and QRS complex in ECG respectively represents :-

(1) Atrial repolarization and Ventricular depolarization

(2) Ventricular depolarization and Atrial repolarization

(3) Atrial depolarization and Ventricular depolarization

(4) Atrial depolarization and Ventricular repolarization

164. Find out the incorrect statements :-

(1) Erythrocyte of mammals is biconcave in shape.

(2) Spleen is called graveyard of body

(3) Blood bank of body is spleen

(4) Decrease in RBC count condition is called Anemia

165. Heart, the .....A..... derived origin, is situated in the .....B..... cavity in between the .....C..... slightly tilted to the .....D..... It is protected by a .....E..... walled membranous bag.

	A	B	C	D	E
(1)	Endoderm	Thoracic	Both lungs	Left	Double
(2)	Ectoderm	abdominal	Both gills	Right	Single
(3)	Mesoderm	Thoracic	Both lungs	Left	Double
(4)	Mesoderm	Abdominal	Both lungs	Right	Single

166. Read the statements carefully and choose correct option as answer :

(A) Inspiration occurs when the intra-pulmonary pressure is more than the atmospheric pressure.

(B) Contraction of the diaphragm increases the volume of the thoracic chamber in the antero-posterior axis.

(C) Contraction of the external intercostal muscles increases the volume of the thoracic chamber in the Anterio-posterior axis.

(D) Relaxation of diaphragm and intercostal muscles leads to expiration.

(1) Only B, C and D are correct

(2) Only A and C are correct

(3) Only B and D are correct

(4) All A, B, C, D and E are correct.

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167. **Statement-I :** Neural signal from pneumotaxic center can reduce the duration of inspiration and there by alter the respiratory rate.
Statement-II : In asthma alveolar walls are damaged due to which respiratory surface is increased.

(1) Statement I is correct and statement II is incorrect.

(2) Statement I is incorrect and statement II is correct.

(3) Both statement I and II are correct.

(4) Both statement I and II are incorrect.

168. Carbon dioxide is carried in blood mainly as:

(1) Sodium Carbonate

(2) Sodium Bicarbonate

(3) Potassium Carbonate

(4) Magnesium Carbonate

169. Total percentage of O_2 transported by heaemoglobin or RBC is :

(1) 3% (2) 97%

(3) 49% (4) 25%

170. Which of the following constitute the exchanging part of the respiratory system

(1) external nostrils to trachea

(2) external nostrils to primary bronchioles

(3) Alveoli and alveolar ducts

(4) external nostrils to terminal bronchioles

171. **Assertion :** A minute vessels vasa recta runs parallel to the Henle's loop.
Reason : It play important role in dilution of the urine.

(1) Both assertion and reason are true but reason is not the correct explanation of assertion.

(2) Assertion is true but reason is false.

(3) Assertion is false but reason is true

(4) Both assertion and reason true and reason is the correct explanation of assertion.

172. Match the column-I with column-II and choose correct answer.

	Column-I		Column-II
A	DCT	i	Filtration of blood
B	Counter current mechanism	ii	Special sensitive region
C	JGA	iii	Maintain concentration gradient of medullary interstitium
D	Malpighian body	iv	Site of conditional reabsorption of Na^+ and water

(1) A-ii, B-i, C-iii, D-iv (2) A-iv, B-ii, C-iii, D-i

(3) A-iv, B-iii, C-ii, D-i (4) A-ii, B-iii, C-i, D-iv

173. Glomerular filtrate devoid of :-

(1) Sugar (2) Salts

(3) Proteins (4) Amino Acids

174. Brush border is a characteristic feature of :-

(1) DCT (2) PCT

(3) Loop of Henle (4) Nephron

175. Uricotelism is found in :-

(1) Birds and mammals

(2) Frogs and toads

(3) Birds, insects and reptiles

(4) Fishes and protozoans

176. An adult human has systolic and diastolic pressure as

(1) 80mm Hg and 120mmHg
(2) 120 mmHg and 80mmHg
(3) 50mmHg and 80mmHg
(4) 80mmHg and 80mmHg

177. Blood of which vessel in mammals carries least percentage of urea ?

(1) Dorsal aorta

(2) Renal vein

(3) Renal artery

(4) Posterior Vena cava

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178. Which of the following stage in cardiac cycle is of longest duration :-

(1) Atrial systole

(2) Ventricular diastole

(3) Joint diastole

(4) Atrial diastole

179. Which of the following statement is incorrect about human heart ?

(1) It is Auto excitable

(2) The entire heart is made of cardiac muscle.

(3) Our heart normally beats 70-75 times in a second.

(4) Both (2) and (3)

180. Match the items given in column I with those in column II and select the correct option given below :-

Column-I		Column-II	
a	Semilunar valve	i	Between right atrium and right ventricle
b	Tricuspid valve	ii	Between left atrium and left ventricle.
c	Bicuspid valve	iii	Between left ventricle and aorta

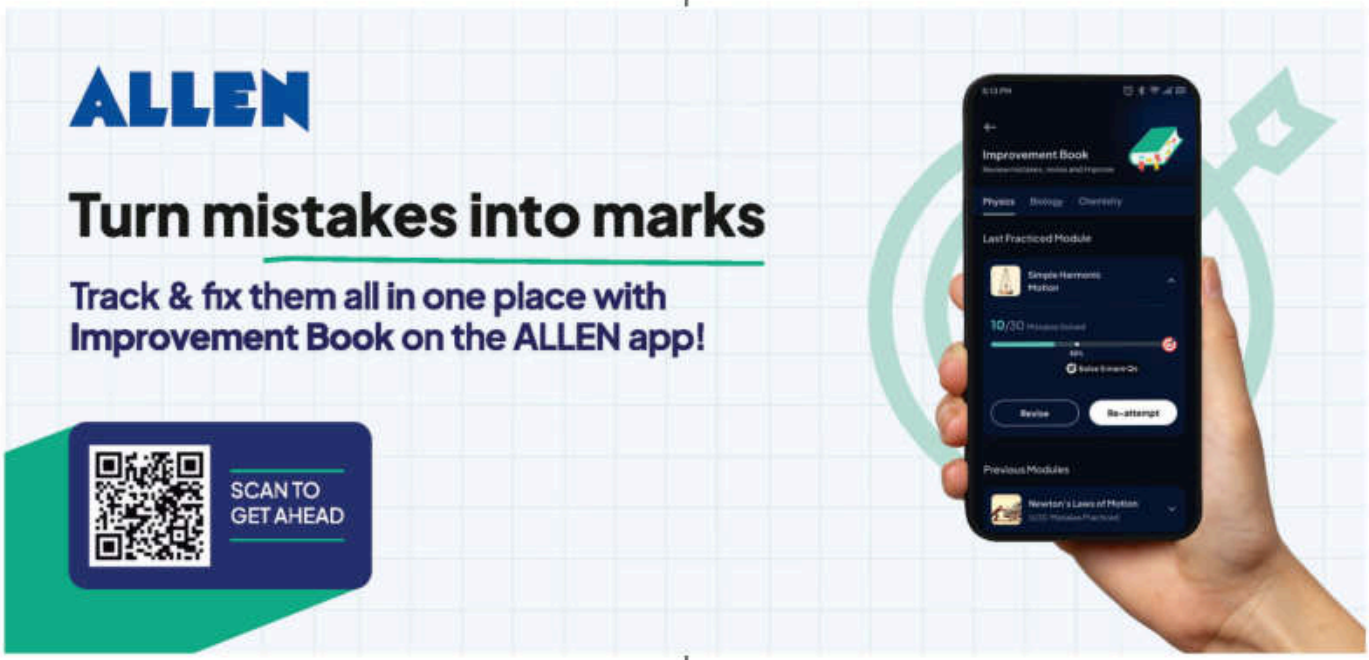
(1) a - iii, b - ii, c - i

(2) a - iii, b - i, c - ii

(3) a - ii, b - i, c - iii

(4) a - ii, b - iii, c - i





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